

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
1	(a)		Diagram showing source, absorber and detector or equivalent stated in words (1) I. Take measurement with no source and no paper/aluminium – [to measure the background radiation] or show awareness of background. II. Measure count rate with no paper/aluminium. (1 for both I and II) III. Insert [a few sheets of] paper between the source and receiver and take measurement. If reduction from count in II then alpha particles present. (1) IV. Insert [a few mm of] aluminium between the source and receiver and take measurement. If reduction in the count from that in III then beta particles are also present and if count is still above the background level, then gamma radiation is present – this penetrates the aluminium. (1)	4			4		4
	(b)	(i)	$\lambda = \frac{\ln 2}{T_{\frac{1}{2}}} = \frac{\ln 2}{(11.4 \times 24 \times 60 \times 60)} \text{ Substitution (1)}$ $7.037 \times 10^{-7} \text{ s}^{-1} \text{ unit (1)}$ Alternative solution: $\lambda = \frac{\ln 2}{T_{\frac{1}{2}}} = \frac{\ln 2}{11.4} \text{ substitution (1) = } 0.0608 \text{ days}^{-1} \text{ unit (1)}$	1	1		2	2	
		(ii)	$\frac{57.0}{11.4} = 5 \text{ so 5 half lives (1)}$ $\text{Activity} = \frac{1}{2^5} A_0 = \frac{1}{32} A_0 \text{ (1)}$ Alternative solution: $A = A_0 e^{-\lambda t}; t = 5T_{\frac{1}{2}} \text{ and } \lambda = \frac{\ln 2}{T_{\frac{1}{2}}}; \text{ substitution } A = A_0 e^{-5 \ln 2} (1)$ $= A_0 \frac{1}{2^5} = \frac{1}{32} A_0 (1)$	1	1		2	2	

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		(iii)	Let the initial number of particles be N_0 , so number of particles remaining after 57 days $= \frac{1}{32} N_0$ i.e. $A \propto N(1)$ Decrease in the number of nuclei in 57 days $= \left(1 - \frac{1}{32}\right) N_0$ i.e. decrease (1) Percentage decrease $= \frac{\left(1 - \frac{1}{32}\right) N_0}{N_0} \times 100\% = 96.875\%$ i.e. percentage (1) Answer only of 3% award 2 marks only	1	1 1		3	3	
			Question 1 total	7	4	0	11	7	4